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SUPERSEDING
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MILITARY SPECIFICATION

ICE CREAM MAKERS; SHAKE MAKERS; AND COMBINATION ICE CREAM AND SHAKE MAKERS - SOFT SERVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This document covers air cooled and sea water cooled (Navy) soft serve ice cream makers; shake makers, and combination ice cream and shake makers.

* 1.2 Classification. The soft serve ice cream makers, shake makers, and combination ice cream and shake makers shall be of the following types, sizes, styles, and classes as specified (see 6.2).

Type I	-	Ice cream makers
Size 1	-	7 gal. per hr. minimum (style A only)
Size 2	-	12 gal. per hr. minimum
Size 3	-	30 gal. per hr. minimum, 15 gal. per hr. each head

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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Type II	-	Shake makers
Size 1	-	15 gal. per hr. minimum (style A only)
Size 2	-	40 gal. per hr. minimum
Size 3	-	64 gal. per hr. minimum - twin head, 32 gal. per hr. each head
Type III	-	Combination ice cream makers and shake maker
Size 1	-	7 gal. per hr. soft serve and 30 gal. per hr. shake minimum
Size 2	-	12 gal. per hr. soft serve and 40 gal. per hr. shake minimum
Style A	-	Counter model (Type I, Size 1 or Type II, Size 1 only)
Style B	-	Floor model
Style C	-	Floor model, caster mounted
Class 1	-	Air cooled
Class 2	-	Sea water cooled (for shipboard use)

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-F-1421	-	Fluorocarbon Refrigerants
FF-C-77	-	Caster, Rigid and Swivel (Institutional Duty)
QQ-C-465	-	Copper-aluminum Alloys (Aluminum Bronze) (Copper Alloy Numbers 606, 614, 630, 632M and 642), Rod, Flat Products with Finished Edges (Flat Wire, Strip and Bar), Shapes and Forgings)
QQ-N-281	-	Nickel-Copper Alloy, Bars, Plates, Rods, Sheets, Strips, Wire Forgings and Structural and Special Shaped Sections
QQ-S-741	-	Steel, Carbon; Structural Shapes; Plates; and Bars

MILITARY

MIL-F-1183	-	Fittings, Tube, Cast Bronze, Silver Brazing
MIL-R-12323	-	Refrigerators and Related Equipment, Packaging and Packing Of

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MIL-R-14328 - Rubber Sheet: Synthetic, Medium Soft, General-Purpose Gasket Material (For Extreme Climatic Conditions)
MIL-C-15726 - Copper-Nickel, Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Rod, and Plate) and Forgings
MIL-B-16541 - Bronze, Valve Castings
MIL-F-20042 - Flange, Pipe and Bulkhead, Bronze (Silver Brazing)
MIL-I-43717 - Ice Milk, Milk Shake, and Yogurt Mixes

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-461 - Electromagnetic Interference and Susceptibility Requirements For the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics Measurement of
MIL-STD-889 - Dissimilar Metals

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity of as directed by the contracting activity.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

UNDERWRITERS LABORATORIES, INC. (UL)

Standard No. 621 - Ice Cream Makers

(Application for copies should be addressed to Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

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NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 6 - Dispensing Freezers

(Application for copies should be addressed to National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167	-	Corrosion-Resisting Chromium Steel Plate, Sheet, and Strip
A 506	-	Specification for Regular Quality, Hot Rolled and Cold Rolled Alloy Steel Sheets and Strips
B 61	-	Steam or Valve Bronze Castings
B 143	-	Tin Bronze and Leaded Tin Bronze Sand Castings
B 148	-	Aluminum Bronze Sand Castings
B 150	-	Aluminum Bronze Rod, Bar, and Shapes
B 280	-	Seamless Copper Tube for Air Conditioning and Refrigeration Field Service

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

* 3.2 Design. The type I, II, and III units shall be capable of making and dispensing the product in ambient temperatures of 75°F and 100°F in the capacities specified in 3.2.1. All material that comes in contact with the product zone shall conform to the material requirements of National Sanitation Foundation Standard No. 6. On all three types of machines, overrun and blending shall be accomplished by means of a gear or belt drive beater/dasher utilizing a gravity flow system. The beater/dasher must whip air into the mix (overrun), scrape frozen mix from the surface of the shell, and move the mass forward in

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the freezer to assist ejection for continuous product draw off. The class 1 shall be air cooled and the class 2 shall be sea water cooled for shipboard use. Style A shall be a counter model, style B shall be a floor model, and style C shall be a floor model with casters conforming to type II, class C, 5 inch diameter, style C, mounting A of FF-C-77. The casters shall be equipped with foot operated brakes. Foot pedal operation shall be provided in addition to the manually operated dispensing mechanism to control the product flow for each dispensing head on the style B machines. No electrical connections or switches shall be installed on the top of any machine. The soft serve ice cream and shake units shall be designed for mix conforming to MIL-I-43717. The frozen product shall conform to requirements of MIL-I-43717. Unless otherwise specified (see 6.2), motors rated up to and including 3/4 horse power (hp) shall be designed to operate on a nominal 120 volt, 60 hertz, (Hz) single phase electrical system and motors rated 1-hp and over shall be designed for operation on a nominal 208 volt, 60 Hz, single phase AC electrical system. When specified for Navy (see 6.2), the ice cream and shake unit shall operate satisfactorily when inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other with no spillage when tested as specified in 4.6.6.

3.2.1 Capacity. Each unit shall be capable of making and dispensing ice cream and shakes as follows when tested as specified in 4.6.1:

TABLE I. Capacity

Type	Size	Ambient temperature ($0^{o}F$)	Ice cream gal. per hr. minimum	Milk shake gal. per hr. minimum
I	1	75	7	-
		100	5	-
I	2	75	12	-
		100	8	-
I	3	75	30 (15 gal. per head)	-
		100	20 (10 gal. per head)	-
II	1	75	-	15
		100	-	15
II	2	75	-	40
		100	-	40

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TABLE I. Capacity (cont'd)

type	Size	Ambient temperature (°F)	Ice cream gal. per hr. minimum	Milk shake gal. per hr. minimum
II	3	75	-	64 (32 each head)
		100	-	64 (32 each head)
III	1	75	7	30
		100	5	30
III	2	75	12	40
		100	8	40

* 3.2.2 Dimensions. Each unit shall not exceed the maximum dimensions specified in table II.

TABLE II. Dimensions, maximum overall in inches

Type	Size	Height	Width	Depth	Style A	Styles B and C
I	1	36	24	32	X	
I	1	60	24	34		X
I	2	60	32	34		X
I	3	60	32	35		X
II	1	36	24	35	X	
II	1	60	24	35		X
II	2	60	32	35		X
II	3	60	32	35		X
III	1	60	32	35		X
III	2	60	32	35		X

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3.3 Standard product. The units delivered under this document shall be the manufacturer's commercial product, except for any changes necessary to comply with document requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

* 3.4 Codes and standards. The soft serve ice cream maker, shaker maker, and combination ice cream and shake maker units shall comply with the applicable requirements of Underwriters Laboratories Standard No. 621, and the applicable requirements of National Sanitation Foundation Standard No. 6 (see 4.5, 4.5.1, and 4.5.2)

* 3.5 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer for soft serve ice cream makers and shake makers provided the completed item complies with all provisions of this document (see 6.6).

* 3.5.1 Copper tube. Copper tube used for refrigeration lines shall conform to ASTM B 280. Tube used for the fabrication of the evaporator and condenser shall be the type normally used for the purpose by the manufacturer except it shall have a wall thickness of not less than 0.016 inch.

* 3.5.2 Bolts, nuts, screws, and washers. Unless otherwise specified in this document bolts, nuts, screws, and washers used for mounting components shall be of the quality and in the sizes, types, and materials normally used for the purpose in commercial practice. Threads shall conform to the FED-STD-H28. Steel bolts, nuts, screws, and washer for the evaporator shall be zinc plated. Brass bolts, nuts, washers, and machine screws shall not used in fastening aluminum parts. Hand operated fastening screws for the freeze head section front plate shall be constructed of 300 series stainless steel, or of hard plastic comparable to polycarbonate in quality. Acrylic type plastic shall not be used.

3.5.3 Insulation. Insulation shall be non-moisture absorbing material; shall be homogenous throughout; and shall have "K" factors specified below (see 3.5.3.1 and 3.5.3.2) based on 70°F mean ambient temperature, and expressed in terms of British Thermal Units per hour per square foot (BTU/hr.ft^2) at a temperature gradient of 1°F per inch of thickness. Loose fill type insulation will not be acceptable. The insulation shall be of a type that is not combustible, and will not support combustion or release toxic fumes or vapors when exposed to high temperatures.

3.5.3.1 Pre-expanded insulation. Pre-expanded insulation shall be polystyrene or polyurethane material having a manufacturer's commercially rated "K" factor of not greater than 0.20 for polystyrene or not greater than 0.15 for polyurethane. The density of the pre-expanded insulation shall not be less than 1.7 pounds per cubic foot when tested as specified in 4.4.1.1.

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3.5.3.2 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial "K" factor of not less than 0.15, and a density of not less than 1.7 pounds per cubic foot when tested as specified in 4.4.1.1.

* 3.5.4 Refrigerant. The refrigerant shall conform to BB-F-1421 type 12, 22, or 502.

3.5.5 Steel.

* 3.5.5.1 Steel, corrosion-resisting. Corrosion-resisting steel shall conform to any of the 300 series of ASTM A 167, or the 400 series of ASTM 176, where permitted by NSF 6.

* 3.5.5.2 Steel, low carbon. Steel, alloy, low carbon shall conform to ASTM A 506.

3.5.5.3 Structural steel. Structural steel shall conform to QQ-S-741.

* 3.6 Construction. The soft serve ice cream maker, shake maker, and combination ice cream and shake maker shall consist essentially of a structural frame unit inclosing a condensing unit, freezer head assembly, mix tank, and all controls necessary for automatic operation. The framework and base shall be fabricated from low carbon steel specified in 3.5.5.2. The exterior panels shall be fabricated of corrosion-resisting steel specified in 3.5.5.1. The exterior panels of the cabinet including the top and front panel, shall each be fabricated from a single sheet. The top and front panels shall have no separations or perforations between the outside edges of the panels except the necessary cutouts for the mix tank on the top panel and the dispensing heads on the front panel. A drip tray, equipped with removable louvers or perforated plates, shall be provided below the dispensing heads of all machines. The freeze head assembly and mix tank shall be insulated with one of the insulations specified in 3.5.3. All component parts that will cause vibration shall be mounted in a manner that will minimize vibration. All accessories, instruments, controls, and tubing shall be fastened or braced. All refrigerant components shall be gas tight when tested as specified in 4.6.2. Contact areas between dissimilar metals shall be insulated to minimize galvanic corrosion by coating the surface of both metals with zinc chromate or asphaltic type compounds or by strips of rubber conforming to MIL-R-14328, except the freezer head bearings. Dissimilar metals are defined by MIL-STD-889. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without removing any other major component. When specified for shipboard use (see 6.2), the ice cream and shake units shall be provided with four mounting holes in the base. The minimum mounting bolt sizes, using four standard bolts, shall be 1/2-inch bolts for equipment not over 400 pounds, and 5/8-inch mounting bolts for equipment not over 700 pounds.

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3.6.1 Insulation installation. All space within the walls of the mix tank and freezer head coil shall be completely filled with pre-expanded insulation specified in 3.5.3.1 or foamed-in-place insulation specified in 3.5.3.2. Pre-expanded insulation, when used, if installed in layers shall have the layers bonded to one another with a minimum of joints, carefully filled and staggered. Insulation shall be bonded to the inner surfaces of the inner liner and outer shell by a commercial grade permanent cement. When copper tubing is used for the evaporator, voids between the tubing shall be filled with a thermosetting mastic.

* 3.6.2 Freezer head section. The freezer head section for single head units shall contain the freezing cylinder, mix tank including lid, mix feed valve, rear head, front head, and removable freezer head door. Twin head units shall contain two each of the aforementioned components. The freezer head section front plate shall be fastened with hardware as specified in 3.5.2.

* 3.6.2.1 Rear head. The rear head, if separate, shall be secured and sealed to the freezer shell. The rear head shall contain the whipper and dasher drive seal, sealed drive bearings, when used, and the dasher drive system.

3.6.2.2. Front head. The front head shall be secured and sealed to the freezer head. The front head shall have provisions for mounting a removable door as specified in 3.6.2.

* 3.6.2.3 Removable freezer head door. The removable freezer head door shall be provided with a sealing arrangement so as to be leak proof when closed. The door shall be held in a closed position by hand operated fasteners.

3.6.2.4 Dasher drive. Connection of the electric motor with the dasher drive system shall be by means of a gear box or one or more V-belts. Provisions for belt adjustment shall be made by moving the electric motor by means of compression or tension bolts, self-adjustment belts, or other self adjusting mechanisms.

* 3.6.2.5 Syrup dispensers. When specified (see 6.2), syrup dispensers in the quantity specified shall be furnished for each ice cream maker and shake maker dispenser. The syrup dispensers shall be refrigerated and be an integral part of the unit located on the front of the unit.

3.6.2.6 Spinners. When specified (see 6.2), spinners shall be furnished and attached to the front of each ice cream maker and shake maker dispenser.

3.6.2.7 Twist lever. When specified (see 6.2), for type I, size 3 only, a twist lever shall be furnished.

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* 3.6.2.8 Wash kit. When specified (see 6.2), a wash kit shall be furnished and ready to attach to a water supply and drain. The wash kit shall consist of, but not be limited to, a chrome plated hot and cold water faucet with swivel mounting and an antisplash aerator, hot and cold copper water lines, a cam operated stainless steel connecting device for attaching the drain hose to the serving spout or a permanently connected drain hose, and a sight glass in the drain hose. Unless otherwise specified (see 6.2), the size of the water inlet supply and drain shall be the manufacturer's commercial model standard.

* 3.6.3 Refrigeration system. The refrigeration system shall be self-contained within the framework. The refrigeration system shall consist of an electric motor driven refrigeration compressor, condenser, controls, and all other accessories, necessary to operate as a completely automatic refrigeration system. The system shall operate from a power source of the voltage phase, and frequency specified (see 3.6.4 and 3.6.5). The use of a transformer to obtain the specified voltage for the condensing unit is prohibited.

3.6.3.1 Compressor. The compressor shall be hermetic or semihermetic type, thermally or mechanically sealed and designed to meet the requirements of 3.2 using the refrigerant specified in 3.5.4.

3.6.3.2 Dehydration and charging. The completely assembled refrigeration system shall be evacuated, dehydrated, and charged with the refrigerant specified in 3.5.4 and operational quantity of refrigerant oil recommended by the manufacturer.

3.6.3.3 Moisture and liquid indicator. A moisture and liquid indicator shall be installed in the liquid line. It shall indicate dry when the refrigeration system is tested as specified in 4.6.3.

* 3.6.3.4 Sea water condenser. The sea water condenser shall be designed for use with sea water at an entering temperature of 85°F and not more than 110°F condensing temperature. Water velocities in the condenser shall not exceed 6 feet per second at design conditions. There shall be no compressor motor overload or loss of condensing capacity with sea water entering the condenser at 90°F. The condenser shall be a type in which the tubes can be mechanically cleaned and shall be installed so that the tubes may be cleaned without removing the condenser from the equipment. All parts in contact with sea water shall be made of copper-nickel (90-10) conforming to MIL-C-15726, aluminum bronze conforming to QQ-C-465, ASTM B 143 copper alloy 903, ASTM B 148 aluminum bronze 955, ASTM B 150 copper alloy 630, or valve bronze conforming to ASTM B 61 except the condenser sea water tubes shall be copper-nickel (90-10) conforming to MIL-C-15726. The heads may be homopolymer acetal resin, or a similar material satisfactory to NAVSE. The condenser shall be provided with specific devices to purge both the refrigerant and water circuits and for draining the water circuit. A water regulating valve shall be provided with each condenser. The

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valve shall be of the direct-acting or pilot-controlled type actuated by condenser gas pressure to modulate flow of water required for the condenser. The valve shall be selected to regulate the water flow from shut-off to required capacity within a maximum operating gas pressure rise of 40 pounds per square inch (psi). The valve shall be adjustable within an operating range of 90 to 140 psi refrigerant gas pressure for R-12, or equivalent where other refrigerants are used. The capacity of the water regulating valve shall at least equal the water requirement of the condenser, based on available inlet water pressure of 35 psi and a pressure drop across the valve of not more than 10 psi. The valve shall be constructed of nonferrous or corrosion-resisting material. The valve body shall be made of valve bronze conforming to MIL-B-16541. In addition, the internal metal parts subject to corrosion or erosion shall be made of nickel-copper conforming to QQ-N-281. The valve shall have either flanged connections conforming to MIL-F-20042 or union ends conforming to MIL-F-1183. The strainer body shall be valve bronze ASTM B 61, 90-10 copper-nickel, monel or other materials satisfactory to the using or procuring activity. The valve shall be constructed in such a manner as to prevent the possibility of entry of sea water to the refrigerant system in the event of derangement. A strainer of No. 20 mesh nickel-copper screen of substantial area shall be provided for the protection of the valve. The strainer basket shall be readily removable for cleaning without disassembly of connecting piping. The strainer shall be not less than 3/4-inch internal pipe size (ips). All sea water piping shall be 90-10 copper-nickel. All sea water fittings shall be 90-10 copper-nickel, valve bronze ASTM B 61 or monel.

3.6.3.5 Brazing and soldering. Brazing or soldering of water lines, refrigeration lines, or component shall be performed to the manufacturer's standard commercial practice. Food contact areas shall conform to NSF No. 6.

3.6.4 Electrical requirements. Unless otherwise specified (see 6.2), electrical components shall be rated for operation from a nominal 115 volt, single phase, 60 Hz power source, or a nominal 208 volt, single phase, 60 Hz power source, or from a combination of both power sources. Transformers, when supplied shall not be used to step up or, step down the voltage for end item use. Transformers shall not be used where specifically prohibited, but may be used as part of the control system (see 3.6.5). When operation from a 50 Hz power source is specified (see 6.2), all motors shall be rated for that frequency.

3.6.5 Controls. Controls shall be designed for operation from a nominal 115 volt, single phase, 60 Hz power source, or a nominal 208 volt, single phase, 60 Hz power source (see 3.6.4). Controls rated as 115 volt, single phase, 60 Hz may be used with a 50 Hz power source, provided that the power can be obtained from a hotline and neutral, or through a transformer (see 3.6.4). Relays, transformers, or any other component necessary for a complete control system shall be furnished as an integral part of the end item.

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3.7 Finish. The finish shall be the manufacturer's standard finish provid-ed it meets NSF requirements. All sharp burrs, slivers, and rough edges shall be removed.

* 3.8 Electromagnetic compatibility. When specified (see 6.2), the soft serve ice cream makers or shake makers shall be designed and equipped for electro-magnetic compatibility in accordance with part 10, group 1 of MIL-STD-461 (see 4.6.4).

* 3.9 Data-name plates. The soft serve ice cream maker or shake maker unit shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for; (a) Methods of applying, (b) Identification tags, (c) Information not required and (d) Optional marking information shall not apply. The data plates shall be made of minimum 20 gauge corrosion-resisting metal and attached to the unit by rivets, screws or welding in such a manner as to meet the applicable National Sanitation Founda-tion sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved or applied by photo-sensitive means:

National Stock Number
Procurement Instrument Identification Number
Specification Data
Manufacturer's Name, Address and Telephone Number
Supplier's Name, Address and Telephone Number (List only if different from Manufacturer)
Manufacturer's Model Number
Government Approved Manual Number (See 6.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the unit.

3.10 Instruction plate. An instruction plate shall be made of corrosion-resisting metal and shall be attached to each ice cream maker, shake maker, or combination ice cream-shake maker. The instruction plate shall bear instruc-tions for start-up, operation, shut-down and preventative maintenance. Also present on the plate shall be the units wiring diagram, refrigeration cycle diagram, type of refrigerant supplied with the system, and any other information required by UL No. 621.

3.11 Workmanship. Sheet metal or structural sections used in fabrication of the soft serve ice cream makers or shake makers shall not be kinked or bent. Bolts, nuts, and screws shall be drawn tight.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required, the ice cream maker and shake unit shall be examined for the defects listed in table III, for dimensions specified in table II and tested in accordance with 4.6.1 through 4.6.3 and when required 4.6.4 and 4.6.6. The presence of any defects listed in table III, any dimensions not within specified requirements, or failure to pass any of the tests in 4.6.1 through 4.6.3 and when specified 4.6.4 and 4.6.6 shall be cause for rejection of the first article (see 6.3).

4.4 Quality conformance inspection. Unless otherwise specified herein, sampling for inspection shall be performed in accordance with provisions set forth in MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.1.1 Density of insulation. Prior to the beginning of each day's production of ice cream makers, shake makers, or combination ice cream-shake makers, one sample of the insulation to be used in that day's production shall be tested as specified in 4.6.5 for conformance to the applicable density requirement in 3.5.3.1 or 3.5.3.2.

* 4.4.2 In-process examination. In-process examination shall be made of the following fabrication operations to establish that no deviations are made from the specified requirements. Whenever nonconformance is noted, correction shall be made to operation and all items affected.

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a. Brass hardware not used to fasten aluminum parts (see 3.5.2)
b. Accessories, instruments, controls, and tubing securely fastened or braced (see 3.6)
c. Contact area between dissimilar metals (see 3.6)
d. Installation of insulation (see 3.6.1)
e. Evaporator (voids between tubes) is filled with thermosetting mastic, when required (see 3.6.1)

* 4.4.3 End item visual examination. The end item shall be examined for the defects listed in table III. The lot size shall be expressed in units of end items of one type only. The sample unit shall be one end item. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE III. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not manufacturer's standard finish, does not meet NSF requirements	X	
	Burrs, rough edges, or slivers	X	
Design, construction and workmanship (applicable to all components and assemblies)	Not design specified	X	
	Any specified component missing or not as specified	X	
	Any component fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, malformed, or otherwise impaired	X	
Freezer head door	Not held in position by hand operated fasteners	X	
Freezer head front plate and fastening screws	Missing or not constructed of material specified	X	
Dasher motor	Adjustment bolts missing (when required), not enough adjustment	X	
Data-name plates	Omitted or not as specified		X
	Information incomplete or not legible		X
	Not located so as to be readily visible		X

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* 4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in 3.2.2. Any dimension greater than the specified maximum shall be classified as a defect. The lot size shall be expressed in units of end items of one type only. The sample unit shall be one end item. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.4.5 End item testing.

4.4.5.1 Refrigerant leakage and moisture testing. Every end item shall be tested as specified in 4.6.2 and 4.6.3. Any end item exhibiting one or more test failures shall be rejected.

4.4.5.2 Representative item testing. One representative end item shall be tested as specified in 4.6.1 and, when specified, in 4.6.4 and 4.6.6 (see 6.5). Any test failure shall be cause for rejection of the lot.

* 4.4.6 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of MIL-R-12323.

* 4.5 Certification compliance examination. Certifications, certified test reports, or listing marks for codes and standards, as applicable, submitted in accordance with 3.4, shall be examined and validated as proof of compliance (see 4.5.1 and 4.5.2).

4.5.1 Underwriters Laboratories, Inc. (UL). Acceptable evidence of meeting the applicable requirements of the Underwriters Laboratories standards shall be a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the soft serve ice cream maker, shake maker, or combination ice cream and shake maker has been tested and conforms to the applicable requirements of the specified UL standards.

4.5.2 National Sanitation Foundation (NSF). Acceptable evidence of meeting the applicable requirements of the National Sanitation Foundation standards shall be one of the following:

a. A listing in the current edition of the National Sanitation Foundation "Listing of Food Service Equipment", and display of the NSF seal on the finished unit.

b. A certification for the unit issued by NSF under their special one-time contract evaluation/certification service.

c. A certified test report acceptable to the contracting officer with the advice of the Army Surgeon General, from an independent testing laboratory, indicating that the unit has been tested and conforms to the specified NSF standards.

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* 4.6 Methods of inspection.

* 4.6.1 Capacity test. The soft serve ice cream maker, shake maker, or combination ice cream maker and shake maker shall be tested for capacity as follows to determine compliance with 3.2:

(1) The test unit shall be at equilibrium in a minimum ambient temperature of 75°F before the start of the test. This minimum ambient temperature shall be maintained during the test. The test item shall be started and the mix tank filled with mix cooled to 40°F. The unit shall be run for 30 minutes or until the product temperature reaches 20°F for soft serve or 28°F for shake. Additional precooled (40°F) mix may be added as required for the remainder of the test. The unit shall operate continuously without overload cutoff during the pull-down of the mix temperature to draw off temperature specified in MIL-I-43717. Then the product shall be drawn off for a period of 1-hour to determine the capacity requirements of 3.2.1.

(2) After completion of part (1), the ambient temperature shall be raised to a minimum of 100°F. When the test unit reaches equilibrium, part (1) shall be repeated except the draw off rate and capacity shall conform to the requirements specified for 100°F in 3.2.1.

(3) Failure of the test unit to make and dispense soft serve or shake in the condition and capacities specified in 3.2.1 shall constitute failure of this test.

4.6.2 Refrigerant leak test. The test item shall be refrigerant leak tested to determine conformance with 3.6. The test for leakage shall be made with a halide leak detector or electronic leak detector or any other method standard with the manufacturer. Any leakage shall constitute failure of this test.

4.6.3 Refrigerant moisture test. The moisture and liquid indicator shall be checked for dryness in conformance to 3.6.3.3 as shown by the component manufacturer's color chart data. A reading indicating moisture shall constitute failure of this test.

* 4.6.4 Electromagnetic compatibility test. When electromagnetic compatibility is required, (see 3.8) the preproduction sample or initial unit of production, as applicable, shall be tested by the contractor in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting office the electromagnetic test report and written certification that the sample unit meets the requirements of the Interference Control Plan, the EMI/EMC Test Plan, and part 10, group 1 of MIL-STD-461.

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4.6.5 Density test for insulation.

4.6.5.1 Free rise method. The sample shall be a 6-inch cube cut from the center of a free foamed cube of insulation measuring a minimum of 8 inches on a side. The 6 inch cube shall be weighed and the density shall be calculated as follows:

$$\text{Density (lbs./ft.}^3\text{)} = 8 \times \text{Weight of 6-inch cube (lbs.)}$$

Any nonconformance with the above shall constitute failure of this test.

4.6.5.2 In-place method. The weight and interior volume of an ice cream maker panel without the foam insulation in place shall be determined. The panel shall then be foamed and weighed. The insulation density shall be calculated as follows:

$$\text{Density (lbs./ft.}^3\text{)} = \frac{\text{Weight of panel after foaming - Weight before foaming}}{\text{Interior volume of panel}}$$

Any nonconformance with the above shall constitute failure of this test.

* 4.6.6 Roll, list, and trim test. When specified for Navy shipboard use (see 6.2), the ice cream and shaker unit shall be bolted to a table or platform. The unit shall be started and the table or platform inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other and operated for 1 hour in each plane. During and after both 1 hour periods, the unit shall be examined to determine compliance with 3.2. Failure of the unit to continue operating or any leaks or malfunctions of any components or any spillage shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

* 5.1 Levels A, B, and Commercial. Preservation, packaging, packing and marking for each level of protection shall be in accordance with the applicable requirements of MIL-R-12323 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use. The soft serve ice cream makers, shake makers, and combination ice cream and shake makers are intended for use in military dining facilities.

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* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type, size, style, and class required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. When procurement is for Navy shipboard (see 3.2 and 3.6).
e. If electrical characteristics are other than specified (see 3.2, 3.6.4 and 3.6.5).
f. When syrup dispensers are required (see 3.6.2.5).
g. When spinners are required (see 3.6.2.6).
h. When twist lever is required (see 3.6.2.7).
i. When a wash kit is required and when the size of water drain connections is other than the manufacturer's commercial model standard (see 3.6.2.8).
j. When electromagnetic compatibility is specified (see 3.8).
k. Selection of applicable level of preservation, packaging, and packing (see 5.1).
l. Government approved manual number to be included or data name plate (see 3.9).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample, consisting of one ice cream maker, shake maker, or combination ice cream and shake maker. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

* 6.4 Changes from the previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in the notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 End item tests. Performance of tests specified in 4.6.1 and, when specified, 4.6.4 and 4.6.6, should be limited to one successful demonstration of compliance with the requirements of this document for each contract.

* 6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.5).

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Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - MD, TS
Navy - MC
Air Force - 82
DGSC - GS

Preparing activity:

Army - GL

Project No. 4110-0387

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-43705D	Ice Cream Makers; Shake Makers; And Combination Ice Cream And Shake Makers - Soft Serve

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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